

Course: Fundamentals of Generative AI and Large Language Models: Theory and Practice

Week 1: Fundamentals of Deep Learning for Generative AI

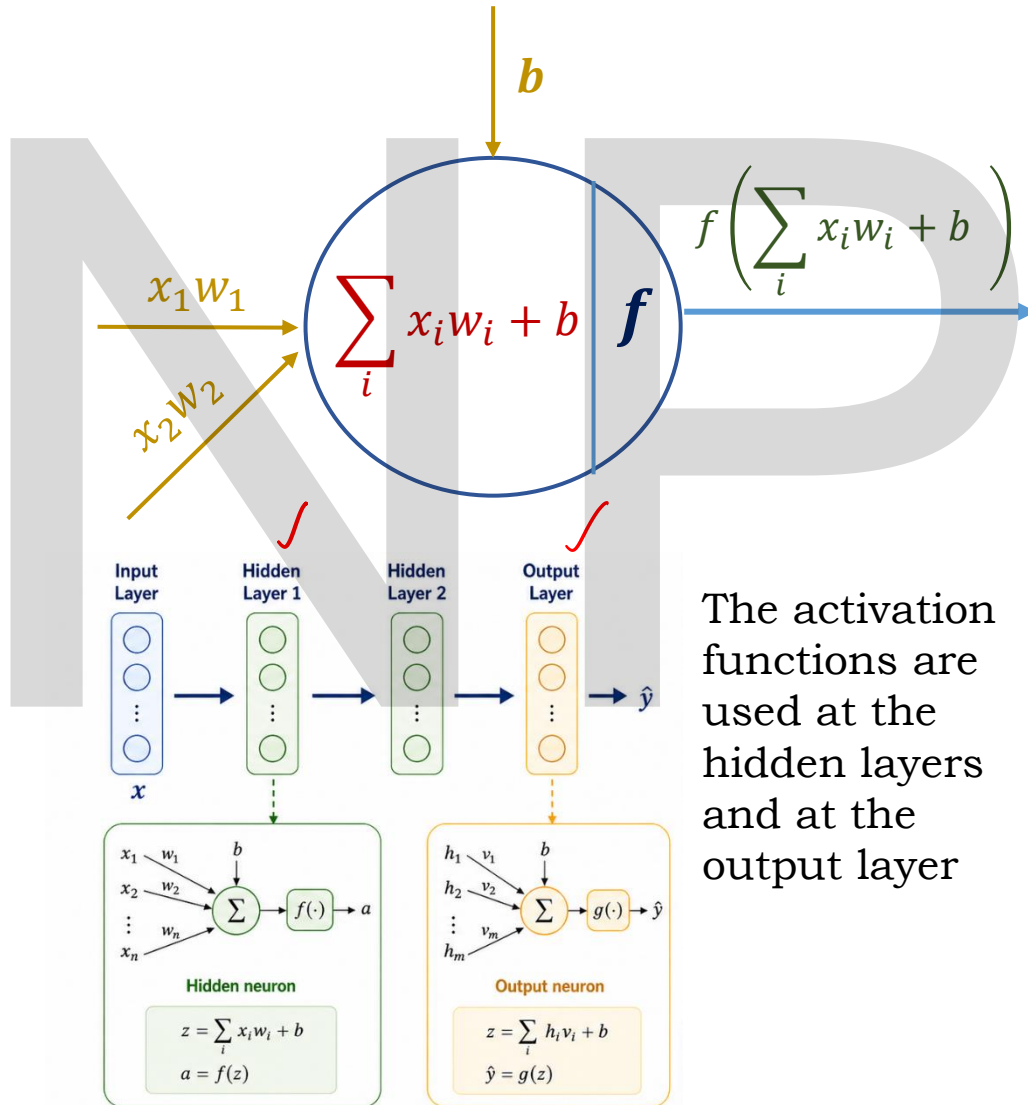
Today's Session Overview

Deep Learning Foundations for Generative AI

- Activation functions
- Loss functions

NLPTEL

Why Activation Functions Are Needed in Deep Networks



The activation functions are used at the hidden layers and at the output layer

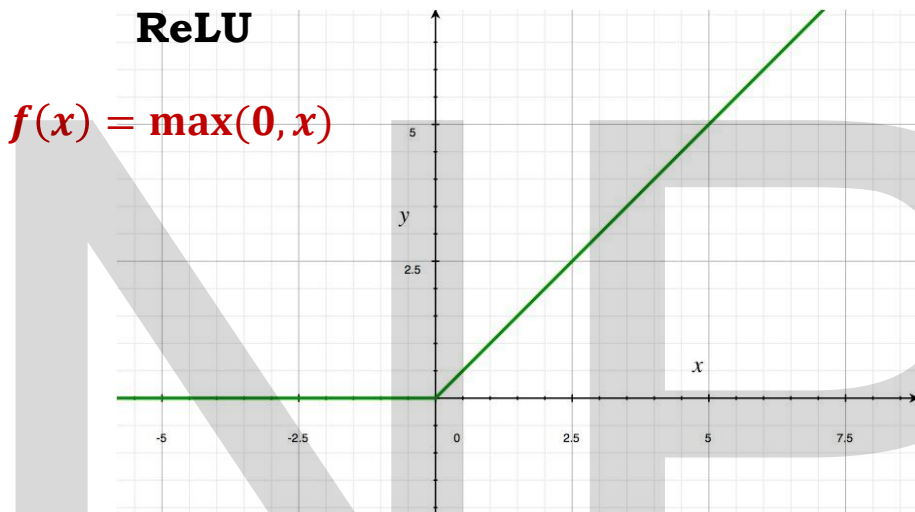
Without an activation function, the weighted sum of inputs is directly passed as the output. This makes the neuron behave like a linear model. Therefore, an activation function **is used to introduce non-linearity into the network. Those activation functions are non-linear activation functions.**

If there are 5 neurons in a hidden layer, each neuron may learn different information from the input. The outputs of these neurons after activation function are combined in the next layer, helping the network learn more complex patterns. **Helps the network learn edges, textures, shapes, words, tokens, and semantic features.**

Activation functions such as ReLU and its variants pass useful positive signals forward and suppress less useful signals, such as negative values. This helps the network focus on important features.

In GenAI, activations help **transform latent/noisy/internal representations into meaningful outputs.**

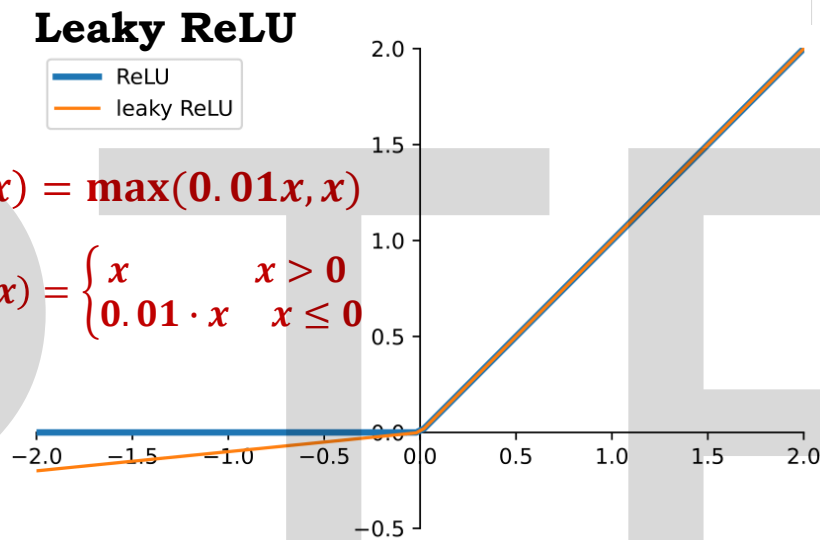
Activation Functions @ Hidden Layer



x	$\max(0, x)$
-3	$\max(0, -3) = 0$
0	$\max(0, 0) = 0$
4	$\max(0, 4) = 4$

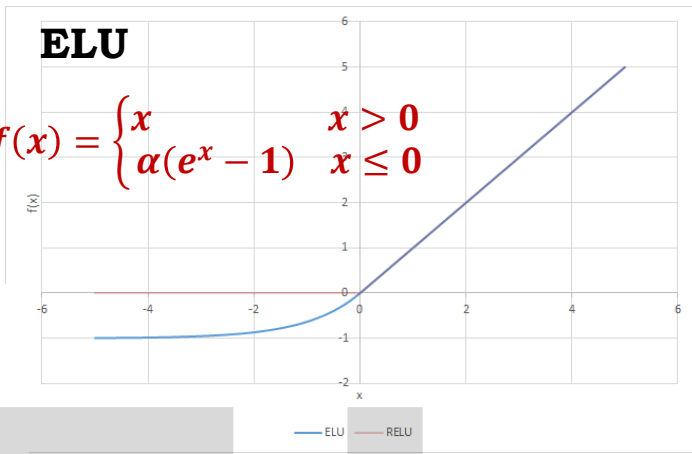
SiLU / Swish:

$$SiLU(x) = x \cdot \sigma(x) = x \cdot \frac{1}{1 + e^{-x}} = \frac{x}{1 + e^{-x}}$$

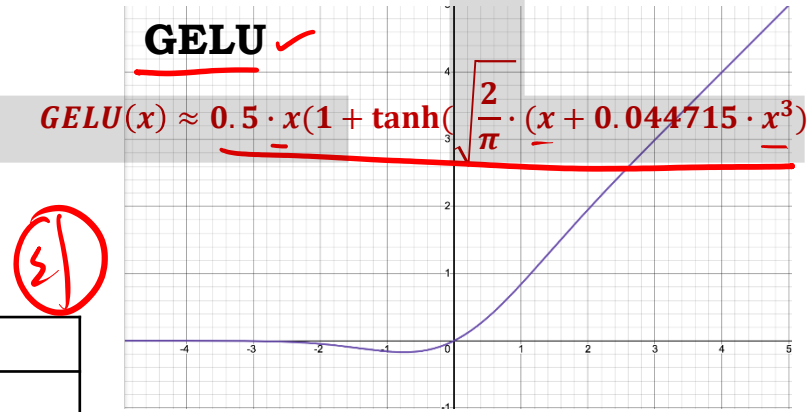


x	$\max(0.01x, x)$
-3	$0.01(-3) = -0.03$
0	$0.01(0) = 0$
4	4

x	GELU output
-1	-0.1587
0	0
1	0.8413
2	1.9545



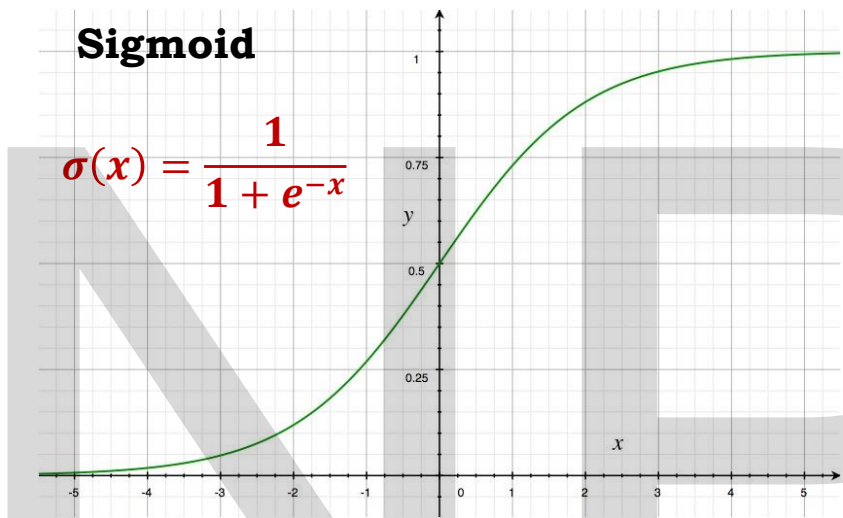
α	$\alpha(e^x - 1)$
-1	$1(e^{-1} - 1) = -0.6321$
2	2



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Activation Functions @ Output Layer

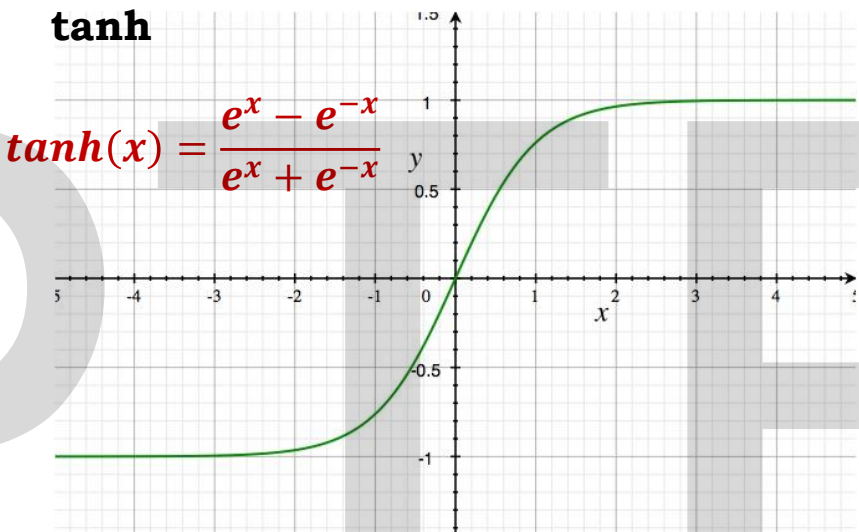
c_1 c_2 c_3 c_4
 c_5



For $x = -2$:

$$\begin{aligned}\sigma(-2) &= \frac{1}{1 + e^2} \\ &= \frac{1}{1 + 7.389} = 0.119\end{aligned}$$

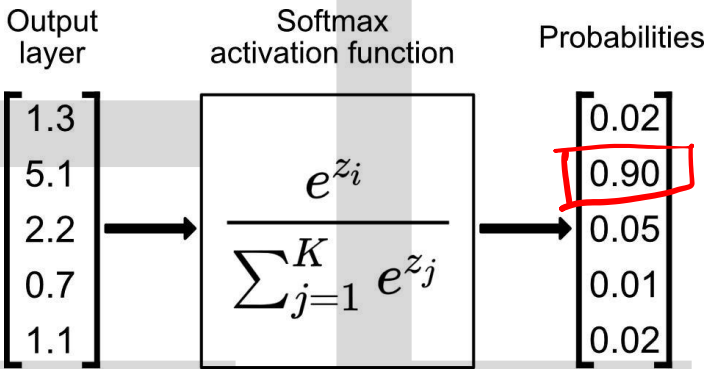
x	Sigmoid output
-2	0.119
0	0.5
2	0.881



x	Tanh output
-2	-0.964
0	0
2	0.964

Softmax:

$$s(z_i) = \frac{e^{z_i}}{\sum_{j=1}^K e^{z_j}}$$



Sigmoid	Binary classification, output between 0 and 1
Tanh	Binary classification when output is represented as -1 and 1
Softmax	Multiclass classification

Where Are Different Activation Functions Used?

Hidden Activation	Where it is commonly used
ReLU, Leaky ReLU/PReLU/ELU ✓	CNNs, Autoencoders, in VAE, in GAN
GELU ✓ and its variants such as GeGLU (Gaussian Error Gated Linear Unit)	Widely used in Transformers and in LLMs
<u>SiLU / Swish</u>	Diffusion Models

Output Activations	Where it is commonly used
Sigmoid/Tanh ✓	Autoencoders, VAE Decoders, GAN Generator
Sigmoid ✓	GAN Discriminator
Softmax ✓	- Used for multi-class classification - Very important in language modeling and next-token prediction in LLMs

CNN

AE

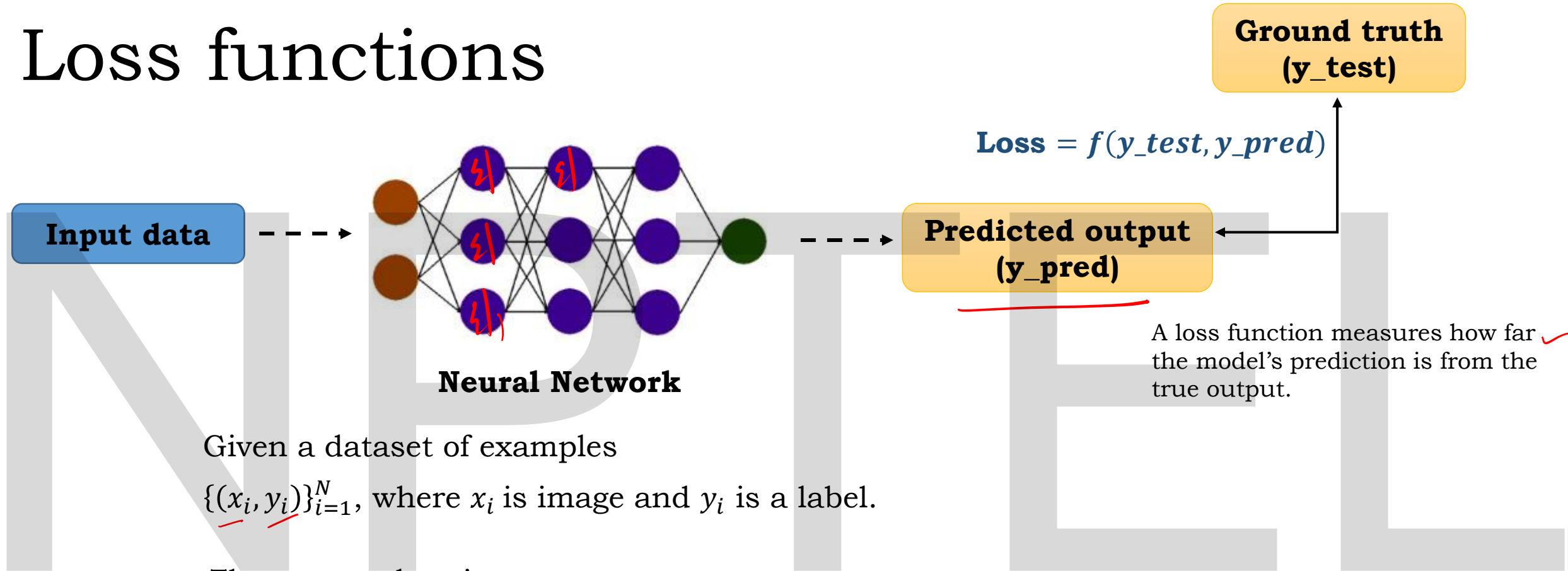
VAE

GAN

D.M
S.M

LLM

Loss functions



Given a dataset of examples $\{(x_i, y_i)\}_{i=1}^N$, where x_i is image and y_i is a label.

The average loss is:

$$L = \frac{1}{N} \sum_{i=1}^N \ell(\hat{y}_i, y_i)$$

where:

$$\hat{y}_i = f(x_i; W) + b$$

The goal of training is to minimize this loss by updating the model parameters W .

In deep learning, the model learns by comparing its predicted output with the expected output. The loss function gives a numerical error value that tells the model how far it is from the desired result.

In **Generative AI**, the objective is not always simple classification. The model may learn to:

- reconstruct the input, as in autoencoders and VAEs; ✓
- generate realistic new samples, as in GANs and diffusion models; ✓
- predict the next token, as in LLMs;
- classify or evaluate generated outputs, when needed.

Therefore, losses in GenAI and LLMs can be broadly understood in two ways:

1. Reconstruction and Classification Losses

2. Generative Model Losses

Reconstruction and Classification Losses

Use reconstruction losses when the target is the input itself; use classification losses when the target is a class or token.

RECONSTRUCTION

Mean Squared Error (MSE)

continuous

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

Measures squared difference between the original and reconstructed value. **Mean Squared Error (MSE)**: used when reconstructing continuous values, such as pixel intensities in an autoencoder.

Binary Cross-Entropy (BCE)

binary

$$L_{BCE} = -\frac{1}{N} \sum_{i=1}^N x_i \cdot \log \hat{x}_i + (1 - x_i) \cdot \log(1 - \hat{x}_i)$$

Binary Cross-Entropy (BCE): used when reconstructing binary or normalized values in the range 0 to 1, such as binary image reconstruction or the original GAN discriminator objective.

CLASSIFICATION

Categorical Cross-Entropy

one-hot label

$$L_{CE} = - \sum_{k=1}^K y_k \log(\hat{y}_k) \quad \checkmark$$

For one-hot labels: Where y_k is the one-hot encoded true label and $\hat{y}_k$ is the softmax probability for the class k

LLMs use cross-entropy loss for next-token prediction

Sparse Categorical Cross-Entropy

integer label

$$L_{SCE} = -\log(\hat{y}_c)$$

Same objective as categorical cross-entropy, but the correct class is stored as an integer index c .

LLMs use sparse-entropy loss for next-token prediction with integer labels

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is Deep Learning

It treats next-token prediction as a **classification problem over the entire vocabulary**.

Loss Functions in Generative Models

KL Divergence

Loss / objective

$$D_{KL}(q_{\phi}(z|x) \parallel p(z))$$

Purpose

Measures how far the encoder distribution is from the prior distribution.

In VAEs, it regularizes the latent space.

VAE

Loss / objective

$$L_{VAE} = L_{reconstruction} + D_{KL}(q_{\phi}(z|x) \parallel p(z))$$

Purpose

Balances input reconstruction with latent-space regularization, usually toward a standard normal prior.

GAN

Loss / objective

L_{GAN} : *G tries to fool D, D tries to identify real/fake* ✓

Purpose

Generator tries to fool the discriminator; Discriminator tries to distinguish real samples from generated samples.

Diffusion

Loss / objective

Noise prediction loss ✓

Purpose

Learns to predict the noise added during the forward diffusion process, enabling denoising-based generation.

Next session

Deep Learning Foundations for Generative AI

- Optimizers for training deep neural networks

NPTREL

NIPTELL

Thank you